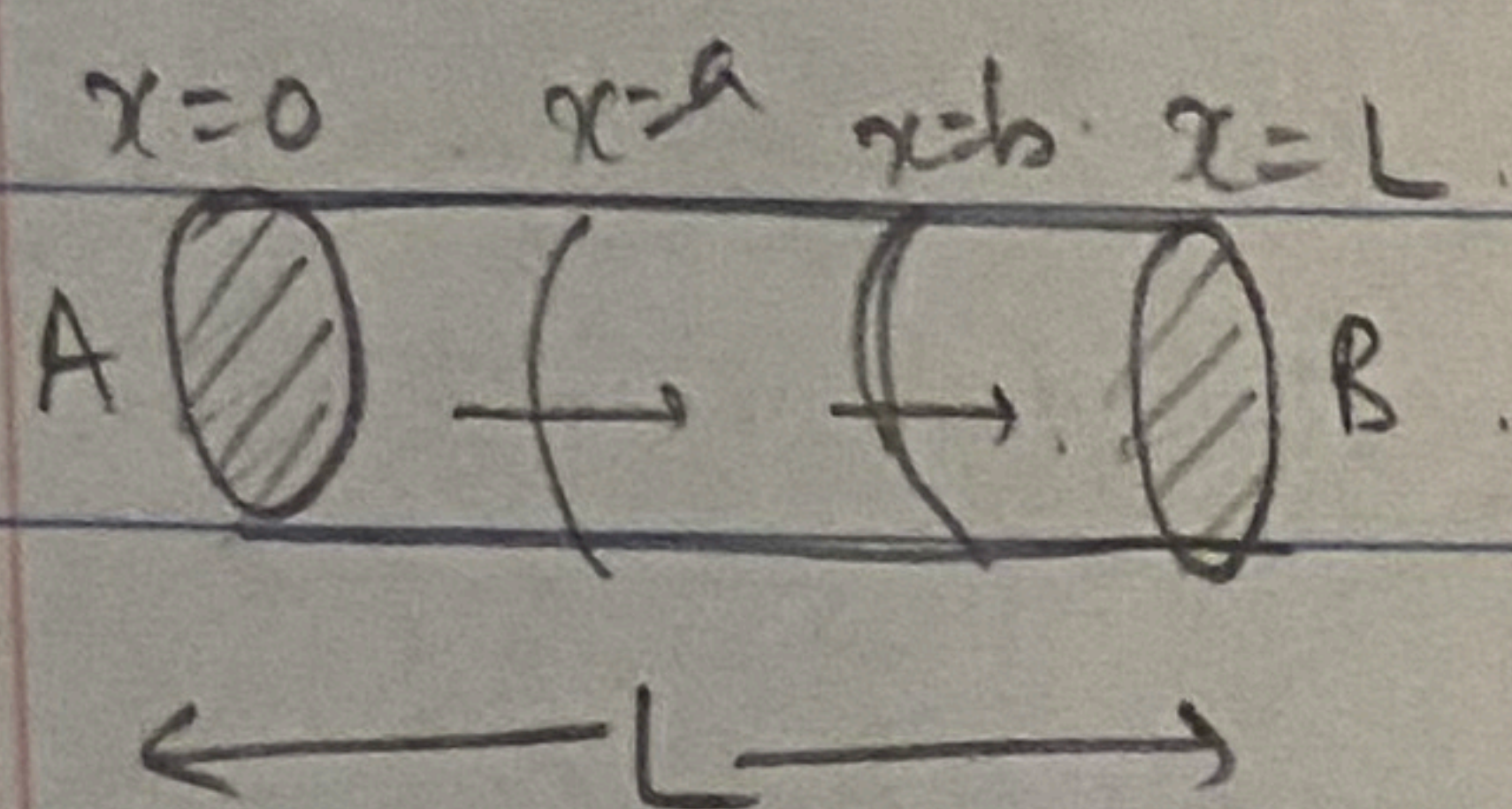


Heat Equation (derivation)



Physical quantities involved -

- temperature - a measure of the warmth/coldness of matter wrt some std value

Denote the temp of the rod at location 'x' time 't' by $u(t, x)$.

- Thermal/heat energy - energy it takes to raise temp. of unit vol. from reference temp 0° to its actual temp. $u(t, x)$. Denote this by

$$e(t, x) \equiv \frac{\text{energy}}{\text{volume}}$$

- specific heat $c(x)$: energy supplied to unit mass of a material to raise its temp. by one unit

$$c(x) \equiv \frac{\text{energy}}{\text{mass} \times \text{temp.}}$$

- mass density - mass per unit vol.

$$\rho(x) \equiv \frac{\text{mass}}{\text{vol.}}$$

$$e(t, x) = c(x) \rho(x) u(t, x) \quad (I)$$

- thermal/heat energy = $\int_a^b e(t, x) \underbrace{A dx}_{\text{vol. element}}$

- heat flux - amount of thermal energy flowing across bodies per unit surf. area per unit time

$$\phi(t, x) \equiv \frac{\text{energy}}{\text{area} \times \text{time}}$$

- Heat source - heat gen. per unit vol per unit time.

$$Q(t, x) = \frac{\text{energy}}{\text{vol} \times \text{time}}$$

- conserv. of heat energy - thermal energy in the slice can only change in time due to heat flowing across the edges and/or being gen. inside.

rate of change of heat energy in time
 = heat energy flowing across slice per unit time
 + heat gen. inside per unit time

$$\frac{d}{dt} \int_a^b e(t, x) A dx = \overbrace{A \phi(t, a)}^{\text{energy in}} - \overbrace{A \phi(t, b)}^{\text{energy out}} + A \int_a^b Q(t, x) dx$$

$$\Rightarrow \underbrace{\phi(t, a) - \phi(t, b)} = \int_a^b \left[\frac{\partial e}{\partial t} - Q \right] dx$$

Fund. thm of calc.

$$- \int_a^b \frac{\partial \phi}{\partial x} dx = \int_a^b \left[\frac{\partial e}{\partial t} - Q \right] dx$$

$$\Rightarrow \frac{\partial e}{\partial t} = - \frac{\partial \phi}{\partial x} + Q \quad (\text{II}).$$

Qualitative ptics of heat cond

- Heat energy flows from hotter to colder reg.
Greater the temp. diff, greater the flow of

$$\Rightarrow \phi(t, x) \propto - \frac{\partial u}{\partial x} \longrightarrow (1)$$

- Diff. materials conduct heat differently. This is quantified by the thermal conductivity $K_0(x)$.

$$\Rightarrow \phi(t, x) \propto K_0(x) \longrightarrow (2)$$

$$(1) \& (2) : \phi(t, x) = -K_0(x) \frac{\partial u}{\partial x}(t, x) \longrightarrow (III)$$

Fourier's law of heat cond

(I) + (II) + (III) :

$$c(x) \rho(x) \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0(x) \frac{\partial u}{\partial x} \right) + Q$$

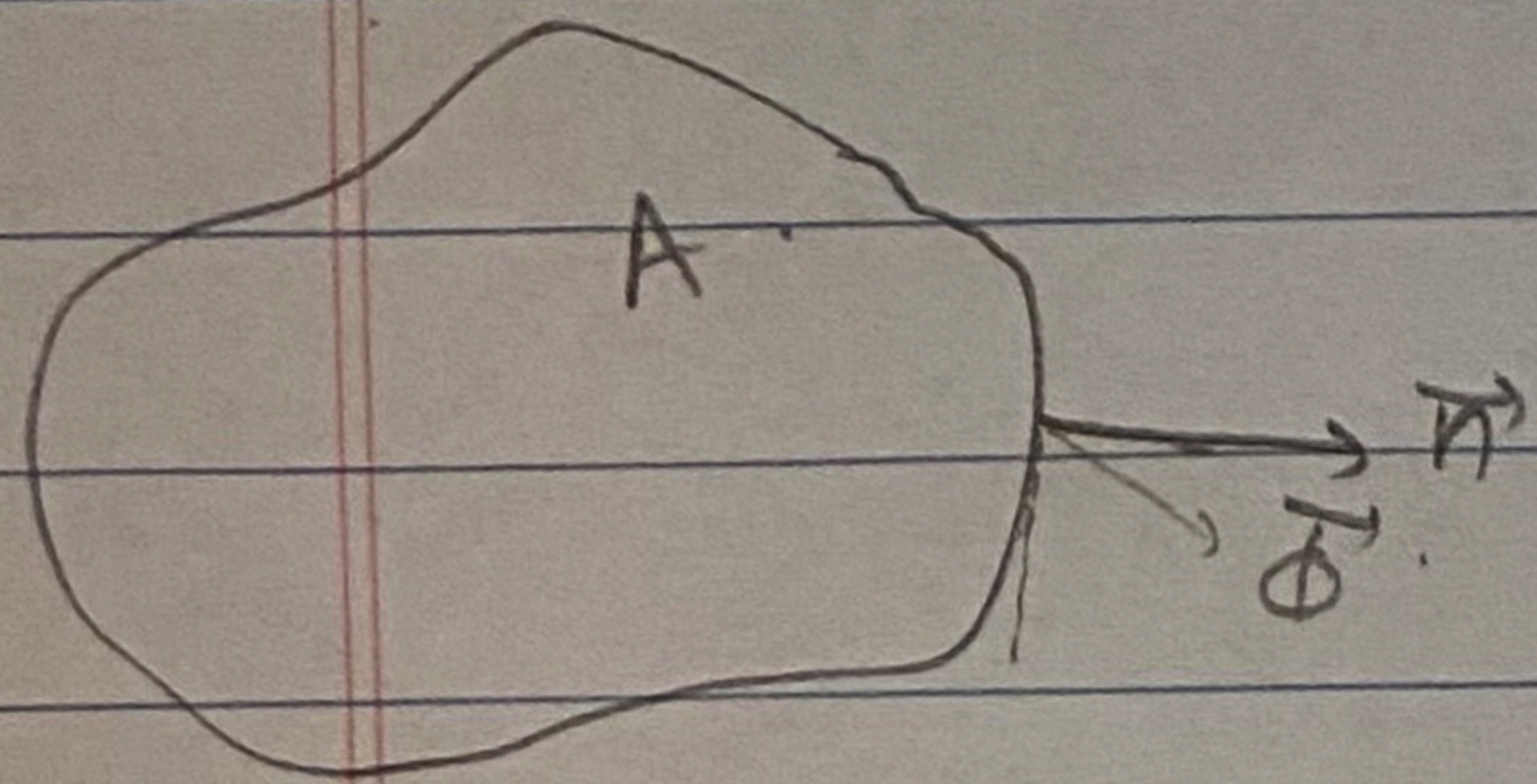
Heat eqn

If c, ρ, K_0 are const & $Q=0$, denoting $K =$

$$\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}$$

Derivation in 2D/3D

$$x \in R \subset \mathbb{R}^n, n=2 \text{ or } 3.$$



Cons. of heat energy:

rate of change of heat energy / time

= heat energy flowing across bdr / time

+ heat energy gen. inside / time

$$\bullet \text{ heat energy} = \int_R c(t, x) \rho(x) u(t, x) dV = \int_R c(x) \rho(x) u(t, x) dV$$

$$\bullet \text{ heat energy gen} = \int_R Q(t, x) dV$$

- heat energy flowing out of R / unit time

$$= \oint_{\partial R} \vec{\Phi} \cdot \vec{n} \, dS$$

$\vec{n}$ unit outward normal vector

[only the normal comp. of heat flow contrib. to heat flowing across bdr]

$\Rightarrow$ cons. of heat energy gives

$$\frac{d}{dt} \int_R c \rho u \, dV = - \underbrace{\oint_{\partial R} \vec{\Phi} \cdot \vec{n} \, dS}_{\text{use div. thm}} + \int_R Q \, dV$$

use div. thm

$$\oint_{\partial R} \vec{\Phi} \cdot \vec{n} \, dS = \int_R \nabla \cdot \vec{\Phi} \, dV$$

change to a vol. int.

$$\Rightarrow \frac{d}{dt} \int_R c \rho u \, dV = - \int_R \nabla \cdot \vec{\Phi} \, dV + \int_R Q \, dV$$

as R was arbitrary,

$$c \rho \frac{\partial u}{\partial t} = - \nabla \cdot \vec{\Phi} + Q$$

Fourier's law: $\vec{\Phi} = -K_0(x) \nabla u$

$$\Rightarrow c \rho \frac{\partial u}{\partial t} = \nabla \cdot (K_0 \nabla u) + Q \quad [\text{heat eq}]$$